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Statistical Zero Knowledge and Quantum One-way Functions

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Prologue

- Fundamental task of cryptography: secure encryption of information against malicious parties (One-way functions)
- Theoretical applications: bit commitment and oblivious transfer, Zero Knowledge Proof Systems and pseudo-random generators
- Candidate one-way functions: Factoring, Discrete Logarithm, Graph Isomorphism, Quadratic Residuosity, approximate Shortest Vector and Closest Vector and the RSA function

Emergence of Quantum Computation

- Possibility of unconditionally secure key distribution
- Classical one-way functions and hence cryptosystems, including RSA, will not be secure against quantum adversaries

Motivations

- Constructing cryptosystems secure even against quantum attacks
- Candidates for quantum one-way functions
- Quantum analogues of the "classic" results on 1way functions

Previous Works

- Quantum computationally secure bit commitment *DumaisMS00, CrépeauLS01, AdcockC01*
- Digital signature schemes (Classical inputs and quantum outputs) *GottesmanC01, Yuen00, LuF04*
- Testing the one-wayness based on the efficiency of constructing a family of reflection operators *KashefiNV02,KawachiKKP04*

Summary

- Quantum Sampling $\longrightarrow$ quantum 1way functions
- Quantum distributionally 1way $\longrightarrow$ quantum 1way functions
- Candidate functions:
  — $\mathcal{L} \in SZK \setminus AvgBQP$
  — $\mathcal{L} \in QMA$-Intermediate

Average case quantum one-way function

Definition

A one to one function $f : \{0,1\}^n \to \{0,1\}^m$ where:

- easy to compute: f can be computed by a polynomial size classical network
- hard to invert: $\exists$ a polynomial $p(\cdot)$ such that $\forall$ poly. time quantum algorithm I and all sufficiently large $n \in \mathbf{N}$,

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} Prob[I(f(x)) \notin f^{-1}(f(x))] > \frac{1}{p(n)}$$

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$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} Prob[I(f(x)) \in f^{-1}(f(x))] \leq 1 - \frac{1}{p(n)}$$

Probability of success

- Closeness of the outcome of I and the outcome of the perfect inverter P, where

$$P : |f(x)\rangle|0\rangle \mapsto |f(x)\rangle|x\rangle$$

- For the case of one-to-one functions

$$\begin{aligned} Prob[I(f(x)) \in f^{-1}(f(x))] &= Prob[I(f(x)) = x] \\ &= |(I(|f(x)\rangle|0\rangle)\ ,\ |f(x)\rangle|x\rangle)|^2 \end{aligned}$$

Equivalent Definition

To demonstrate the relation between quantum one-way functions and the Quantum Sampling:

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Definition

...

- hard to invert: $\exists$ a polynomial $p(\cdot)$ such that $\nexists$ poly. time quantum algorithm I' that for all sufficiently large $n \in \mathbf{N}$:

$$I' : |f(x)\rangle|0\rangle \mapsto a_{f(x)}|f(x)\rangle|x\rangle + b_{f(x)}|f(x)\rangle|G_{f(x)}\rangle \qquad (1)$$

where $G_{f(x)}$ is a garbage state, $\frac{1}{2^n}\sum_{x\in\{0,1\}^n} a^2_{f(x)} \geq 1 - \frac{1}{p(n)}$ and $a_{f(x)}$ are positive real numbers.

Proof

- By copying the input in an entangled register we make the garbage state orthogonal to the correct state
- By running twice the inverting algorithm and assuming, without loss of generality, that the initial amplitudes were real numbers, we can make the amplitudes positive real numbers

Quantum State Generation

- To prepare efficiently a certain superposition state
- We restrict ourselves to superpositions that correspond to samplable classical probability distributions

Quantum Sampling of a classical circuit C

- D_C be the distribution over outputs of C
- $|C\rangle = \sum_{z\in\{0,1\}^m} \sqrt{D_C(z)}|z\rangle$

Definition (Aharonov and Ta-shma)

CQS problem: For a given C find a quantum circuit of size poly($|C|$) which prepares the quantum sample $|C\rangle$:

$$|QS_C(|0\rangle) - |C\rangle| \leq \epsilon$$

CQS and Quantum one-way functions

Theorem

Assume for a classical circuit C, which computes a one-to-one function, the corresponding CQS problem is hard, then the function $f_C : x \mapsto C(x)$ is a quantum one-way function.

Proof

From C we obtain $U_f : |x\rangle|b\rangle \mapsto |x\rangle|f(x) \oplus b\rangle$

Assume f is not one-way: $\forall p \exists I'$ which inverts f approximately:

$$I' : |f(x)\rangle|0\rangle \mapsto a_{f(x)}|f(x)\rangle|x\rangle + b_{f(x)}|f(x)\rangle|G_{f(x)}\rangle$$

where $|G_{f(x)}\rangle$ is a garbage state and $\frac{1}{2^n}\sum_x a^2_{f(x)} > 1 - \frac{1}{p(n)}$.

we can construct the following circuit QS_C:

$$\begin{aligned} |x\rangle|0\rangle \quad & \Rightarrow_{U_f} |x\rangle|f(x)\rangle \\ & \Rightarrow_{SWAP} |f(x)\rangle|x\rangle \\ & \Rightarrow_{I'} a_{f(x)}|f(x)\rangle|0\rangle + b_{f(x)}|f(x)\rangle|G_{f(x)}\rangle \end{aligned}$$

Starting with a uniform superposition of $x \in \{0,1\}^n$ we have

$$\frac{1}{2^{n/2}} \sum_{x\in\{0,1\}^n} |x\rangle|0\rangle \Rightarrow_{QS_C}$$

$$\frac{1}{2^{n/2}} \sum_{x} \left(a_{f(x)}|f(x)\rangle|0\rangle + b_{f(x)}|f(x)\rangle|G'_{f(x)}\rangle \right) \equiv Q$$

Let $|C\rangle = \frac{1}{2^{n/2}} \sum_x |f(x)\rangle|0\rangle$ be the quantum sample of the circuit

$$\begin{aligned} |(Q\ ,\ |C\rangle)|^2 &= |\frac{1}{2^n} \sum_x a_{f(x)}|^2 \\ &\geq |\frac{1}{2^n} \sum_x a^2_{f(x)}|^2 \\ &> (1 - 1/p(n))^2 \\ &> 1 - \epsilon \end{aligned}$$

Many-to-one Case

Our result extends to the many-to-one functions with the assumption that the inverting algorithm for the one-way function provides a uniform superposition of the preimages of f.

Quantum distributionally one-way function

Definition

A many-to-one function $f : \{0,1\}^* \longrightarrow \{0,1\}^*$ where:

- easy to compute: f can be computed by a polynomial size classical network.
- hard to sample: There exists a polynomial $p(\cdot)$ such that for any quantum polynomial time algorithm S and all sufficiently large $n \in \mathbf{N}$,

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} |(S(|f(x)\rangle|0\rangle)\ ,\ |f(x)\rangle|H_{f(x)}\rangle)|^2 \leq 1 - \frac{1}{p(n)},$$

where $|H_{f(x)}\rangle = \frac{1}{\sqrt{|f^{-1}(f(x))|}} \sum_{x \in f^{-1}(f(x))} |x\rangle$.

QSC and Quantum Distributionally one-way functions

Theorem

Assume for a classical circuit C, which computes a many-to-one function, the corresponding CQS problem is hard , i.e. there exists no poly($|C|$) size quantum circuit implementing QS_C. Then the function $f_C : x \mapsto C(x)$ is a quantum distributionally one-way function.

Sampling vs Inverting

Theorem (Luby and Impagliazzo)

if $\exists$ a distributional one-way function then $\exists$ a general one-way function.

Sampling vs Inverting

Theorem (Luby and Impagliazzo)

if $\exists$ a distributional one-way function then $\exists$ a general one-way function.

- Assume f is dist. 1way and we know the size of the preimage, $k = \lfloor \log |f^{-1}(f(x))| \rfloor + O(\log n)$ Then the following function is 1way:

$$g(x, h_k) = (f(x), h_k, h_k(x))$$

where h_k is a random hash function $h_k : \{0,1\}^n \longrightarrow \{0,1\}^k$.

Quantum case

Theorem

if $\exists$ a quantum distributional one-way function then $\exists$ a general quantum one-way function.

Proof

- Assume, we know the $k = \lfloor \log |f^{-1}(f(x))| \rfloor + O(\log n)$
- Picking a random hash function and a random string

$$Q \;:\; |k\rangle|0\rangle \rightarrow |k\rangle \sum_{h_k} |h_k\rangle \quad , \quad B \;:\; |k\rangle|0\rangle \rightarrow |k\rangle \sum_{r_k} |r_k\rangle$$

- Quantum inverter of g as

$$I \;:\; \left\{ \begin{array}{lll} |f(x)\rangle|h_k\rangle|h_k(x)\rangle|0\rangle|0\rangle & \rightarrow_\epsilon & |f(x)\rangle|h_k\rangle|h_k(x)\rangle|x\rangle|0\rangle \\ |f(x)\rangle|h_k\rangle|s_k\rangle|0\rangle|0\rangle & \rightarrow & |f(x)\rangle|h_k\rangle|s_k\rangle|0\rangle|1\rangle \end{array} \right\}$$

$$T : |h_k\rangle|h_k(x)\rangle|x\rangle \quad \rightarrow \quad |h_k\rangle|0\rangle|x\rangle$$

Partial Quantum Sampler PQS(f(x),k)

$$|f(x)\rangle|k\rangle|0\rangle|0\rangle|0\rangle|0\rangle \Rightarrow$$

$$\sqrt{p_k}|f(x)\rangle|k\rangle|0\rangle|0\rangle|H_{f(x)}\rangle|0\rangle + \sqrt{1-p_k}|f(x)\rangle|k\rangle|G_{f(x),k}\rangle|1\rangle$$

Partial Ancilla Preparation, PAP(f(x),k)

$$|f(x)\rangle|k\rangle|0\rangle|0\rangle|0\rangle \Rightarrow$$

$$|f(x)\rangle|k\rangle\Big(p_k|0\rangle + (1-p_k)|1\rangle\Big)|0\rangle + |f(x)\rangle|G_{f,k}\rangle|1\rangle$$

Ancilla Preparation AP(f(x))

$$|f(x)\rangle|n\rangle|0\rangle|n-1\rangle|0\rangle\cdots|1\rangle|0\rangle|0\rangle \Rightarrow$$

$$|f(x)\rangle|n\rangle\cdots|1\rangle\sum_j q_j|j\rangle|0\rangle + |G_f\rangle|1\rangle$$

where $q_j = \prod_{i=1}^{j-1}(1-p_i)p_j$

Quantum Sampler, QS(f(x))

$$|f(x)\rangle|0\rangle|0\rangle \Rightarrow$$

$$\sum_j q_j^2 \sqrt{p_j}|f(x)\rangle|H_{f(x)}\rangle|0\rangle + |G_{f(x)}\rangle|1\rangle$$

Candidates

- Our main theorem provides a framework for seeking candidates
- Quantum states hard to generate $\Rightarrow$ the intractability of inversion
- Classically samplable distribution $\Rightarrow$ the ability to generate instances together with the auxiliary information

Statistical Zero Knowledge

- *SZK* proof: To prove an statement without yielding anything beyond its validity.
- *SZK* language: There is a *SZK* interactive proof system to decide it, (verifier can learn only about $x \in L$).

- (Aharonov Tash-ma 03) Any language $\mathcal{L} \in SZK$ can be reduced to a family of instances of *CQS* problem.

- (Sahai and Vadhan 99) Consider two constants $0 \leq \beta < \alpha \leq 1$ such that $\alpha^2 > \beta$, define $SD_{\alpha,\beta}$ to be the problem of deciding for any two given classical circuits C_0 and C_1 whether:

$$\begin{array}{lcll} \|D_{C_0} - D_{C_1}\| & \geq & \alpha & \text{or} \\ \|D_{C_0} - D_{C_1}\| & \leq & \beta & \end{array}$$

  where it is promised one to occur.

SZK and Quantum one-way

The quantum analogue of Ostrovsky's result:

Theorem

Assume there exists a language $\mathcal{L} \in SZK \setminus AvgBQP$, then quantum one-way functions exist.

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Proof. Assume $\mathcal{L} \in SZK \setminus AvgBQP$ and let C_0 and C_1 be the classical circuits which decide L via reduction to the SV-complete language. Then either of the corresponding functions f_0 or f_1 is quantum one-way.

Conclusions

- Hard instances of q. sampling $\Rightarrow$ q. one-way functions
  Easy instances $\Rightarrow$ efficient quantum algorithms

- $SZK \not\subseteq AvgBQP \Rightarrow$ Hard instances
  What other assumptions $\Rightarrow$ hard instances of the CQS?

- Candidate one-way problems: Graph Non-Isomorphism and approximate Closest Lattice Vector problem
  Can we construct one-way functions from other problems, such as the hidden subgroup problem in the dihedral or other non-abelian groups?

- Watrous proved q. one-way functions $\Rightarrow$ computational zero knowledge for NP
  Other such implications?